Union.

$\alpha \to \beta$

$\alpha \to \gamma$

decomp.

$\alpha \to \beta\gamma$

Dept → Building

Dept → Budget

→ generalized Union.

composition $\alpha \to \beta$

$m \to n$

$\alpha m \to \beta n$

Syllabus:

Indexing + big data analysis.

1. Ref : $\alpha \to \beta$ $\quad \beta \subseteq \alpha$

2. Aug : $\alpha \to \beta \Rightarrow \alpha\gamma \to \beta\gamma$

3. Tran : $\alpha \to \beta$, $\beta \to \gamma \Rightarrow \alpha \to \gamma$

Additional:

1. Union
2. Decomposition
3. Pseudo
4. comp.

$\alpha \to \beta\gamma \Rightarrow \alpha \to \beta$, $\alpha \to \gamma$

$\alpha \to \beta$

$\gamma \to \delta$ $\Rightarrow \alpha\gamma \to \beta\delta$

Ref: CID → CID

CID, Credit → Credit

Aug:

SID, Dept → Budget SID

1, CSE → 120, 1

student No. 1 of CSE maps to 120 budget & SID 1.

Order important!

Trans:

SID → Dept Dept → Budget

SID → BudgetA

SID, CID → Dept, credit

↳ converse NOT true.

2 possibilities.

SID → credit	SID → Dept
CID → Dept	CID → credit

A → BC , B → E , CD → EF

A → B

.

AD → CD

CD → EF

AD → EF

AD → E

AD → F

Part 2

SID	Name	CGPA	Hall

SID → Name, CGPA, Hall

SID → {SID, Name, CGPA, Hall}

→ can determine all the intra records

so it's a superkey.

see whether a candidate key

or not.

Fundamental steps:

Find all possible candidate key.

Cand. key:

α = SK ⟶ already superkey

$\alpha\beta\gamma$ = SK